

Task 2: Predictive Model Design & Implementation Logic

Project: Geldium Finance Credit Card Delinquency Forecasting Architecture

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Section 1: Generated Model Logic & Architecture (Step 1)

Core Model Selection

For Geldium Finance's delinquency forecasting model, an XGBoost (Extreme Gradient Boosting) classifier is chosen as the primary architecture. This decision-tree-based ensemble learning algorithm excels at handling mixed numerical and unstandardized categorical arrays (like the messy Employment_Status values found during EDA) while optimizing non-linear relational patterns. To satisfy internal risk governance, a linear Logistic Regression baseline model will run alongside it to act as a benchmark for interpretability and compliance stability.

End-to-End Predictive Modeling Pipeline

The architecture processes raw consumer credit portfolios through a structured four-stage automated lifecycle:

Data Ingestion & Cleaning → Feature Engineering & Encoding → XGBoost Model Processing → Risk Prediction Matrix

- Data Ingestion & Cleaning Engine:** Imports the source rows, enforces structural datatype changes (mapping Month_1 through Month_6 from text labels like "Late" or "Missed" to numerical ordinal integers 1 and 2), and applies median/zero imputation to missing fields (Loan_Balance, Income).
- Feature Engineering & Encoding:** Automatically standardizes numerical matrices like Credit_Score and converts Employment_Status variations into clean, one-hot encoded variables.
- Model Processing Layer:** Executes gradient boosted decision trees, aggregating weak learners sequentially to minimize loss functions against the target field Delinquent_Account.
- Prediction Output Matrix:** Generates a soft probability decimal value (0.00 to 1.00) for each Customer_ID, mapping directly to risk tranches (Low, Medium, High Risk).

Top 5 Most Impactful Input Features

Feature	Description
Credit_Utilization	The exact percentage of available credit in use; higher utilization rates heavily weight the probability output toward delinquency.
Month_1 – Month_6 (Ordinal)	The historical trailing timeline of late/missed payment milestones acts as a primary operational indicator of default likelihood.

Credit_Score	The historical consumer credit rating which establishes a baseline creditworthiness threshold.
Debt_to_Income_Ratio	Measures a customer's fixed financial overhead liabilities relative to their total incoming cash-flow streams.
Missed_Payments	The cumulative count of missed deadlines over the past 12 months, indicating systemic or chronic financial strain.

Section 2: Model Justification & Business Context (Step 2)

Operational Fit and Trade-off Matrix

The selection of an XGBoost architecture balances high-performance prediction capabilities with realistic operational constraints within financial services. While deep neural networks offer marginally higher fit metrics on vast data vectors, they function as non-transparent "black boxes," lacking the explainability needed for financial audits.

Conversely, standalone linear models often struggle to capture multi-layered risk factors, such as a client with high income but extremely elevated short-term utilization. XGBoost handles missing data organically and maps non-linear correlations with high accuracy.

To bridge the gap between complexity and transparency, we integrate SHAP (SHapley Additive exPlanations). This approach breaks down exactly how much variables — such as a low credit score or missed payment — contribute to an individual customer's final risk classification, ensuring full alignment with regulatory compliance mandates and consumer transparency laws.

Section 3: Model Performance & Fairness Evaluation Framework (Step 3)

To ensure the deployment remains both accurate and mathematically fair across diverse customer demographics, the validation pipeline relies on four core metrics:

1. Primary Performance Metrics

- **Area Under the ROC Curve (AUC-ROC):** Measures how effectively the model distinguishes between non-delinquent (0) and delinquent (1) accounts across all classification thresholds. Target baseline: ≥ 0.85 .
- **F1-Score (Precision vs. Recall):** Because delinquency datasets are structurally imbalanced (fewer delinquent accounts than active ones), evaluating overall accuracy can be misleading. The F1-Score balances precision (minimizing false accusations against reliable customers) with recall (maximizing the identification of true defaults).

2. Fairness & Bias Mitigation Audits

- **Disparate Impact Ratio:** Calculates the selection rate of high-risk flags across sensitive demographic metrics (such as Age cohorts or geographical Location profiles). A fair model must keep this ratio within the strict bounds of 0.80 to 1.25 to prevent discriminatory credit bias.

- **Equalized Odds Compliance:** Ensures that the model's error rates (false positives and false negatives) are statistically uniform across protected groups, so a young self-employed customer has the same baseline error probability as an older corporate employee.